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Gate -2011-IN -16 Question

QUESTION

For the Boolean expression $f = \bar{a}\bar{b}\bar{c} + \bar{a}b\bar{c} + a\bar{b}\bar{c} + abc + ab\bar{c}$, the minimized Product of Sum (PoS) expression is

(A) $f = (b+\bar{c}).(a+\bar{c})$

(B) $f = (\bar{b}+c).(\bar{a}+c)$

(C) $f = (\bar{b}+c).(a+\bar{c})$

(D) $f = \bar{c} + abc$

QUESTION ANALYSIS

Solution

Assuming the Boolean expression is

$$f = \bar{a}b\bar{c} + a\bar{b}\bar{c} + \bar{a}bc + abc + ab\bar{c}$$

Step 1: Convert to Minterms

$$\bar{a}b\bar{c} = m_2$$

$$\bar{a}bc = m_3$$

$$a\bar{b}\bar{c} = m_4$$

$$ab\bar{c} = m_6$$

$$abc = m_7$$

Therefore,

$$f = \Sigma m(2, 3, 4, 6, 7)$$

Step 2: Find the Zero Cells

The missing minterms are

$$m_0,\ m_1,\ m_5$$

Hence,

$$f = \Pi M(0,1,5)$$

Step 3: K-map Grouping of Zeros

AB\C	0	1
00	0	0
01	1	1
11	1	1
10	1	0

Group 1: m_0 **and** m_1 Common variables:

$$a = 0, \qquad b = 0$$

Maxterm:

$$(a + b)$$

Group 2: m_1 **and** m_5 Common variables:

$$b = 0, \qquad c = 1$$

Maxterm:

$$(b + \bar{c})$$

Step 4: Write the Minimized PoS Expression

$$\boxed{f = (a + b)(b + \bar{c})}$$

Verification

$$(a + b)(b + \bar{c})$$

is equal to 0 for

$$000,\ 001,\ 101$$

and equal to 1 for

$$010,\ 011,\ 100,\ 110,\ 111$$

which exactly matches the given function.

$$\boxed{\text{Minimized PoS} = (a + b)(b + \bar{c})}$$

Components Required

S.No	Component	Quantity
1	Pygmy FPGA Board	1
2	Seven-Segment Display	1
3	USB Cable	1
4	Computer/Laptop (Linux/Ubuntu/Termux)	1
5	FPGA Toolchain (QL SymbiFlow)	1
6	Connecting Wires (if external display is used)	As Required

Hardware Connections

Pygmy FPGA Pin	Connected To	Function
16	Input A	Boolean Function Input A
17	Input B	Boolean Function Input B
18	Input C	Boolean Function Input C
20	segment a	Seven-Segment Display Segment a
21	segment b	Seven-Segment Display Segment b
22	segment c	Seven-Segment Display Segment c
23	segment d	Seven-Segment Display Segment d
25	segment e	Seven-Segment Display Segment e
26	segment f	Seven-Segment Display Segment f
27	segment g	Seven-Segment Display Segment g
3.3V	COM Pin	Common Connection of Seven-Segment Display

Note: Pins 16, 17, and 18 are used as inputs corresponding to variables A, B, and C, respectively. Pins 20–27 (excluding pin 24) are connected to the seven-segment display segments a through g. The COM pin of the display is connected to the 3.3V supply.

Execution Procedure

1. Create the Verilog source file `lcd`$_{t}ask.vandtheconstraintfile$

1. `Compile the design using the QL SymbiFlow toolchain`

   ```
   ql_symbiflow -compile \
   -d ql-eos-s3 \
   -P pu64 \
   -v lcd_task.v \
   -t top \
   -p lcd_task.pcf
   ```

2. `After successful compilation, the bitstream file is generated inside the build directory.`

   ```
   build/top.bit
   ```

3. Rename the generated .bit file to a .bin file:

```
mv build/top.bit top.bin
```

4. Copy the generated .bin file to the computer where the FPGA programming environment has been established.

5. Open a terminal and navigate to the directory containing the generated binary file:

```
cd <path_to_binary_file>
```

6. Put the Pygmy FPGA board into Flash Mode and connect it to the computer using a USB cable.

7. Verify that the board is detected by the system:

```
ls -l /dev/ttyACM0
```

   If the device is detected, information about the serial port will be displayed.

8. Program the FPGA board using the TinyFPGA programmer utility:

```
python3 tinyfpga-programmer-gui.py \
--port /dev/ttyACM0 \
--m4app boolean_display_0.bin \
--node n4
```

9. Wait until the programming process completes successfully.

10. Press the RESET button on the Pygmy FPGA board.

11. Observe the output on the hardware and verify the expected functionality.

Results and Discussion

The Verilog design implementing the simplified Boolean function was successfully synthesized, compiled, and programmed onto the Pygmy FPGA board. After loading the generated binary file and resetting the board, the output of the Boolean function was displayed on the seven-segment display.

For every combination of the input variables A, B, and C, the FPGA evaluated the Boolean expression and displayed either `0` or `1` on the seven-segment display. The observed outputs matched the expected theoretical results obtained from the simplified expression.

Figure **??** shows the output observed on the seven-segment display during hardware testing.

The truth table obtained during testing is shown below.

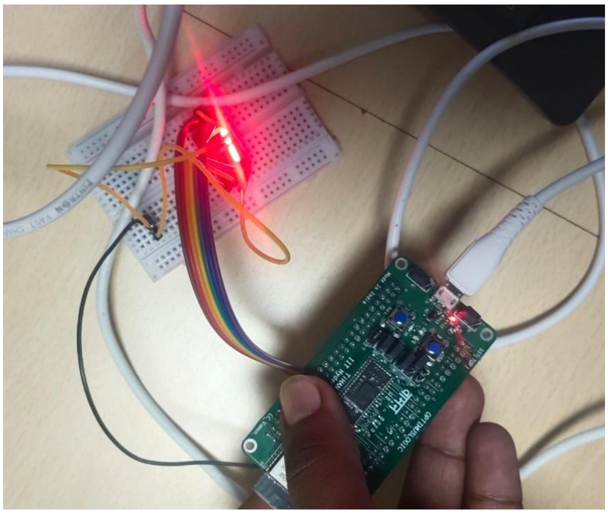

Figure 1: Output Displayed on the Seven-Segment Display

A	**B**	**C**	**Displayed Output**
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1